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User Acceptance of Refinery Valves

API RECOMMENDED PRACTICE 591
SECOND EDITION, FEBRUARY 1998

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Manufacturing, Distribution and Marketing Department

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FOREWORD

This recommended practice is based on the knowledge and experience of petroleum refiners, valve manufacturers, and others, and its objective is to describe practices that will result in a purchaser's receipt of valves which consistently meet API valve specifications.

Any modifications, deletions, and amplifications necessary for individual users should be made by supplementing this recommended practice rather than by rewriting it.

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Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products (Sections 763.160-179).

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User Acceptance of Refinery Valves

1 Scope

This recommended practice describes actions a purchaser can take to procure new valves that are required to be in accordance with the standards listed in Section 2, but the purchaser may find that the suggested actions also apply to valves manufactured to other specifications. This recommended practice does not take the place of a manufacturer's quality control or quality assurance program, and it does not release the manufacturer from the responsibility for providing valves of consistent quality that meet API, ASME, and ANSI standards and any additional requirements of the purchase specification.

2 Referenced Publications

The most recent editions of the following standards, codes, and specifications are cited in this recommended practice:

API

Std 594	*Wafer and Wafer-Lug Check Valves*
Std 598	*Valve Inspection and Testing*
Std 599	*Metal Plug Valves—Flanged and Welding Ends*
Std 600	*Steel Gate Valves—Flanged and Butt-Welding Ends, Bolted and Pressure Seal Bonnets*
Std 602	*Compact Steel Gate Valves—Flanged, Threaded, Welding, and Extended-Body Ends*
Std 603	*Class 150, Cast, Corrosion-Resistant, Flanged-End Gate Valves*
Std 608	*Metal Ball Valves—Flanged, Threaded, and Butt-Welding Ends*
Std 609	*Butterfly Valves: Double Flanged, Lug- and Wafer-Type*

ASME[1]

B1.1	*Unified Inch Screw Threads (UN and UNR Thread Form)*
B16.5	*Pipe Flanges and Flanged Fittings*
B16.10	*Face-to-Face and End-to-End Dimensions of Valves*
B16.11	*Forged Fittings—Socket-Welding and Threaded*
B16.25	*Butt-Welding Ends*
B16.34	*Valves—Flanged, Threaded, and Welding End*
B18.2.2	*Square and Hex Nuts (Inch Series)*
B31.3	*Process Piping*

MSS[2]

SP-55	*Quality Standard for Steel Castings for Valves, Flanges and Fittings and Other Piping Components—Visual Method*
SP-92	*MSS Valve User Guide*

3 Terms and Definitions

3.1 acceptance criteria: Specified limits placed on the characteristics of an item, process, or service defined in codes, standards, or other requirement documents.

3.2 audit: A planned and documented activity performed to determine by investigation, examination, or evaluation of objective evidence the adequacy of and compliance with established procedures, instructions, drawings, and other applicable documents and the effectiveness of implementation. An audit should not be confused with surveillance or inspection activities performed for the purpose of process control or product acceptance.

3.3 characteristic: Any property or attribute of an item, process, or service that is distinct, describable, and measurable.

3.4 corrective action: A measure taken to rectify conditions adverse to quality and, where necessary, to preclude repetition of such conditions.

3.5 document: Any written or pictorial information that describes, defines, specifies, reports, or certifies activities, requirements, procedures, or results.

3.6 guideline: A suggested practice that is not mandatory in programs intended to comply with a standard. The word **should** denotes a guideline; the word **shall** denotes a requirement.

3.7 inspection: An examination or measurement to verify whether an item or activity conforms to specified requirements.

3.8 item: A term used to indicate any of the following: assembly, component, equipment, material, part, structure, subassembly, or unit.

3.9 nonconformance: Any item or action that does not meet the requirements of the standards listed in 2.0, or any additional requirements of the purchase specification or the manufacturer's standard.

3.10 owner: The person, group, company, agency, or corporation who has or will have title to the valve.

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

[2]Manufacturers Standardization Society of the Valve and Fittings Industry, Inc., 127 Park Street, N. E., Vienna, Virginia 22180.

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3.11 procedure: A document that specifies or describes how an activity is to be performed.

3.12 procurement documents: Purchase requisitions, purchase orders, drawings, contracts, specifications, or instructions used to define requirements for purchase.

3.13 qualifications (personnel): Characteristics or abilities gained through education, training, or experience (as measured against established requirements, such as standards or tests) that qualify an individual to perform a required function.

3.14 qualified procedure: An approved procedure that has been demonstrated to meet the specified requirements for its intended purpose.

3.15 quality assurance: Refers to all of the planned and systematic actions necessary to provide confidence that a structure, system, or component will perform satisfactorily in service.

3.16 repair: The process of restoring a nonconforming item to a condition in which the item functions reliably and safely even though it still does not conform to the original requirement.

3.17 right of access: The right of a purchaser, an owner, or a designated representative to enter the premises of a supplier for the purpose of an inspection, a surveillance, or a quality assurance audit.

3.18 supplier: Any individual or organization that furnishes items in accordance with a procurement document. It is a term used to indicate any of the following: vendor, seller, contractor, subcontractor, fabricator, or consultant.

3.19 testing: A process that verifies the capability of an item to meet specified requirements by subjecting it to a set of physical, chemical, environmental, or operating conditions.

3.20 traveler: A process control document which accompanies the work through manufacturing.

4 Purchaser Training

Members of the purchaser's organization who are involved with the procurement of valves, including the person responsible for producing the purchase specification, the purchasing agent, and the person who receives the valves at the job site, must possess a knowledge of valves and understand that the valves must meet appropriate specifications. The purchaser should verify the source of supply through the appropriate manufacturer to assure that the purchased valves are newly manufactured in accordance with the appropriate standards. MSS SP-92 can be used to train the appropriate members of the purchaser's organization.

5 Quality Assurance

The purchaser should determine that the valve manufacturer can supply valves which meet the purchase specifications and that the manufacturer will continue to supply valves of appropriate quality in the future. The purchaser should also verify that the manufacturer has a working quality assurance program. The essential elements of a quality assurance program are described in Section 7. An up-to-date list of authorized distributors should be made available by the manufacturer upon request.

6 Product Verification

To initially qualify a valve manufacturer, the purchaser should, as a minimum, verify that the manufacturer's products meet API, ASME, and ANSI valve standards by examining the design of the valve and verifying service experience with the manufacturer's products. A more extensive review that includes testing samples of the manufacturer's valves is recommended. The recommended evaluation procedure is detailed in Section 8.

7 Guidelines for Quality Assurance Program

7.1 PURPOSE

This section provides guidelines for establishing and executing a quality assurance program for the production of new refinery valves.

7.2 QUALITY ASSURANCE PROGRAM

7.2.1 General

A documented quality assurance program, one that defines the activities and items to which it applies, should be planned, implemented, and maintained. When the program is established, the technical aspects of activities that affect quality should be considered.

7.2.2 Organization

The organizational structure, functional responsibilities, levels of authority, and lines of communication for activities that affect quality should be documented. The persons responsible for establishing and conducting a quality assurance program should have the authority, access to work areas, and organizational freedom to identify quality problems; initiate or recommend solutions to quality problems through designated channels; verify the implementation of the solutions; and when a problem is discovered, assure that further processing, delivery, installation, or use is controlled until proper disposition has been made of the nonconformant, deficient, or unsatisfactory condition.

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7.2.3 Personnel Selection

The personnel selected to perform inspection and testing should have experience or training commensurate with the scope, complexity, or special nature of these activities.

7.3 DESIGN

7.3.1 General

The manufacturer should establish and maintain documented procedures to control and verify the design of the product to assure that specified requirements are met.

7.3.2 Design Input

Applicable design input considerations, such as design bases, performance requirements, and codes and standards, should be identified and documented and their selection reviewed by the manufacturer for adequacy. Incomplete, ambiguous, or conflicting requirements should be resolved with those responsible for imposing the requirements.

7.3.3 Design Review and Verification

At appropriate stages, formal documented design reviews should be conducted and documented by individuals other than those responsible for developing the original design.

Design verification should be conducted and documented to demonstrate conformance of the product to design requirements.

7.3.4 Design Changes

Design changes and changes to design documents should require the same controls as the original design and design documentation.

7.4 DOCUMENT CONTROL

The preparation, distribution, and modification of documents, such as drawings, specifications, and procedures, that specify quality requirements or prescribe activities affecting quality should be controlled to assure that the correct versions of the documents are employed.

7.5 CONTROL OF PURCHASED ITEMS AND SERVICES

7.5.1 General

The purchase of items from suppliers should be controlled by the manufacturer to assure conformance with the specified requirements. This control should provide for the following as appropriate:

a. Evaluation and selection of the source.
b. Evaluation of objective evidence of quality, furnished by the supplier.
c. Inspection, audit, and examination of items and services upon delivery or completion.

7.5.2 Documents

In the procurement documentation, the manufacturer should require that suppliers have a quality assurance program consistent with the program described in this section.

7.5.3 Receiving Inspection

The manufacturer should use receiving inspection to verify the conformance by a supplier to the specified requirements. The manufacturer should establish acceptance criteria for the items inspected. If the procurement documents require the supplier to furnish documentation, the receiving inspection should be coordinated with a review of the supplier's documentation.

7.5.4 Control of Supplier Nonconformances

The manufacturer and supplier should establish and document methods for the disposition of items that do not meet the requirements of the procurement documents.

7.5.5 Right of Access

At each level of procurement, the manufacturer's procurement documents should require the supplier to provide access to his plant facilities and records for inspection or audit by the manufacturer, his designated representative, and other parties authorized by the manufacturer.

7.6 CONTROL OF PROCESSES

7.6.1 General

Processes that affect the quality of items or services should be controlled by instructions, procedures, drawings, checklists, or travelers. Special processes that control or verify quality, such as those used in welding, heat treatment, and nondestructive examination, should be performed by qualified personnel in accordance with the specified requirements.

7.6.2 Acceptance Criteria

The applicable requirements of relevant codes and standards, including acceptance criteria for the processes, should be specified or referenced in the manufacturer's procedures or instructions.

7.6.3 Records

Records should be maintained as appropriate for currently qualified personnel, processes, and the equipment for each special process.

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7.7 INSPECTION

7.7.1 General

Characteristics to be inspected and inspection methods to be employed should be specified. Identification of a rejected item should be maintained until the item is repaired or scrapped. Inspection for acceptance should be performed by persons other than those who performed or directly supervised the work being inspected.

7.7.2 Personnel

Inspection personnel should be supervised and managed by staff who have organizational independence from manufacturing and production personnel.

7.7.3 In-Process Inspection

Where necessary to verify quality, items should be inspected while in process or under construction. If it is impossible or disadvantageous to inspect items while in process or under construction, indirect control should be provided by monitoring of the processing methods, equipment, and personnel. Inspected items should be segregated from uninspected items.

7.7.4 Methods

Inspection methods should be established and executed to verify that the characteristics of an item are within the specified limits. Inspection methods should include the verification of the calibration and integrity of instruments and instrument systems and the verification of maintenance, as appropriate.

7.8 TEST CONTROL

7.8.1 General

The characteristics to be tested and the test methods to be employed should be specified. Identification of a rejected item should be maintained until the item is repaired or scrapped.

7.8.2 Test Requirements

Test requirements and acceptance criteria should be provided or approved by the organization responsible for the design of the item to be tested. Test requirements and acceptance criteria should be based on requirements specified in applicable design or other pertinent technical documents.

7.8.3 Test Procedures

Test procedures should include or reference test objectives and provisions for assuring that the prerequisites for the given test have been met, that adequate instrumentation is available and used, and that necessary monitoring is performed. In lieu of specially prepared written test procedures, appropriate sections of related documents, such as ASTM methods, supplier manuals, equipment maintenance instructions, and approved drawings or travelers with acceptance criteria, can be used.

7.8.4 Test Results

Test results should be documented (for example, by travelers and checklists) and evaluated by a responsible authority to assure that the test requirements have been satisfied. Tested items should be segregated from untested items. Test result documents do not need to be retained.

7.8.5 Equipment

The selection of measuring and test equipment should be controlled to assure that the equipment is of the proper type, range, accuracy, and tolerance.

7.8.6 Calibration

Measuring and test equipment should be calibrated, adjusted, and maintained at the prescribed intervals or, before use, against equipment certified to have valid relationships to nationally recognized standards.

7.9 HANDLING, STORAGE, AND SHIPPING

The handling, storage, and shipping of items should be conducted in accordance with established procedures or specified instructions.

7.10 AUDITS

The manufacturer should perform annual audits to verify compliance with all aspects of the quality assurance program and to determine the program's effectiveness. These audits should be performed in accordance with written procedures or checklists by personnel who do not have direct responsibility for the activities being audited. The audit results should be documented and should be reviewed by the responsible management. Follow-up action should be taken where it is indicated.

7.11 CORRECTIVE AND PREVENTIVE ACTION

7.11.1 General

a. Documented procedures should be established and maintained for implementing corrective and preventive action.

b. Any corrective or preventive action taken to eliminate the causes of actual or potential nonconformances should be to a degree appropriate to the magnitude of the problem and commensurate to the risks encountered.

c. Changes to documented procedures resulting from corrective and preventive action should be implemented and recorded.

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7.11.2 Corrective Action

Corrective action should include:

a. Identifying, investigating, and documenting the cause of repetitive nonconformances related to product, process, and/or quality system, and recording the results.

b. Determining corrective action needed to eliminate cause of nonconformances.

c. Applying controls to assure corrective actions are taken and are effective.

7.11.3 Preventive Action

Preventive action should include:

a. Using appropriate sources of information (e.g., processes, work operations, audit results, quality records, etc.) to detect, analyze and eliminate potential causes of nonconformances.

b. Initiating preventive action and applying controls to assure it is effective.

7.12 STATISTICAL TECHNIQUES

The manufacturer should identify the need for statistical techniques required for verifying the acceptability of process capability and product characteristics. Documented procedures should be established and maintained to implement and control the required statistical techniques.

8 Evaluation of Valves Manufactured to API and ASME Valve Standards

8.1 ENGINEERING ANALYSIS

8.1.1 Drawings, Specifications, and Qualifications

8.1.1.1 For the valve types being evaluated, the valve manufacturer shall provide general assembly drawings of one valve size of each design for each pressure class. These drawings shall clearly show the details of construction, including stem-to-wedge, disk, ball, or plug connection; guides; bearings; stem seals; body joint facing and seal; and seat details. The drawings shall cover the sizes used in the laboratory tests (see 8.1.2.2 through 8.1.2.9) and the calculations (see 8.1.1.6). They shall also include complete descriptions of the construction materials for all of the parts, including fasteners. The specifications for the materials shall be U.S. specifications such as those provided by ASTM. The manufacturer shall also furnish a list of the trim and sealing materials used in the valves.

8.1.1.2 The manufacturer shall furnish a tabulation of body (and bonnet or cover) minimum thicknesses; stem and shaft diameters; packing-gland lengths and stuffing-box depths; pressure ratings; and end connections for all sizes of all valves being evaluated. If material specifications in addition to those described in 8.1.1.1 are involved, they shall be described.

8.1.1.3 The manufacturer shall furnish a tabulation of all tapped openings in valve bodies, bonnets, and covers for all valves included in the qualification.

8.1.1.4 If welding, including casting repair, is used in the manufacture of the valves, the appropriate qualification documents must be available for evaluation.

8.1.1.5 Upon request, the manufacturer shall make available a tabulation of the current sources from which castings and forgings used for bodies, bonnets, covers, and closure elements (for example, wedges, disks or balls) are obtained and all sources from which completed valves are obtained. Sources include licensees who are authorized to manufacture or sell the product under the manufacturer's logo. The tabulation shall identify the sources for each valve size, material, and pressure class.

8.1.1.6 The manufacturer shall make available to the laboratory conducting the strength tests required in 8.1.2.7 a guide for sizing the required test fixtures by providing the calculated stem/shaft-to-closure element failure loads for the valves to be tested. In addition, the manufacturer shall include in the submittal to the purchaser (see 8.4) sample calculations and a tabulation for the theoretical failure loads for all valves in each pressure class.

The calculations shall be presented in a step-by-step format that is complete, clear, and easy to follow. All basic equations and material properties shall be presented, and the sources of the equations and data shall be documented. Sketches must be included to show clearly the failure paths investigated and the nominal dimension for each part (i.e., stem major/minor diameters, threading, undercut, wedge-ears, ball-slot, etc.). For calculation of stem-to-closure element failure loads, the manufacturer's minimum stem dimensions shall be used.

8.1.1.7 The manufacturer shall provide the recommended closure torques or rim-pull/handle forces required to adequately seat the valve for all sizes of valves being evaluated.

8.1.1.8 The manufacturer shall specify the standards used to design bolted joints (body and bonnet) and provide sample calculations on various sizes and pressure classes.

8.1.2 Inspections and Tests

8.1.2.1 The valve manufacturer shall engage an independent testing laboratory, use their own laboratory, or use a combination of these two to perform the inspections, examinations, and tests described in this section. The facilities used shall be mutually agreeable to the purchaser and the manufacturer, and these facilities shall prepare a comprehensive report and evaluation of their test results. The laboratory, or combination of laboratories, shall be professionally competent to

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conduct, or supervise the conduct of, the required tests, and the laboratory staff should include a degreed metallurgical engineer or mechanical engineer. The laboratory, or combination of laboratories, should be equipped to perform, or supervise the performance of, nondestructive examination, physical tests, and chemical analyses on materials. The laboratory staff should be familiar with the API standards referenced in this publication and with the codes, standards, and specifications referenced in those standards. The personnel who conduct the various inspections, examinations, and tests shall have had prior experience with similar testing or shall be instructed by the laboratory professional staff before the test program starts.

The laboratory's proposed program shall be mutually agreeable to the purchaser and the manufacturer to make certain that the program adequately covers the following:

a. Qualifications of the personnel performing the inspections and tests.

b. Test details and format used to present the results of the tests.

c. Number, sizes, and types of valves examined (see Appendix A).

d. Source of the test valves and test valves selection method.

To ensure that the valves were not made specifically for the tests, a random sampling feature shall be incorporated into the program. In the event the manufacturer chooses to use an outside laboratory for this test program, the laboratory personnel shall select the test valves randomly from the manufacturer's or distributor's stock. If the manufacturer chooses to perform all tests using his own laboratory (or a combination of his own and an outside laboratory), the manufacturer shall utilize an outside independent professional firm (e.g., an outside laboratory/auditor) or qualified individual (e.g., a licensed Professional Engineer—not an employee of the manufacturer) to select the test valves randomly from the manufacturer's or distributor's stock. This independent firm or individual shall be acceptable to the purchaser, shall confirm that the inspection and tests required by this document have been performed, and shall sign the final report.

It is expected that the manufacturer shall have sufficient stock from which a random sampling of their valve products may be selected. If sufficient sizes or quantities of the sample lot valves recommended in Appendix A are not available for selection, the sample lot may be as agreed upon by the purchaser and manufacturer. As part of the selection process, the independent party shall document the number, size, class, and type of valves available, and shall clearly identify (by stamping, tagging, etc.) the valves actually selected. After work commences on the agreed upon program, testing shall be completed on the randomly selected sample lot with no substitutions.

8.1.2.2 All of the pressure tests specified in API Standard 598, including the optional closure tests, shall be made on each valve. Seating surfaces shall be dry or coated only with a very light oil no heavier than kerosene. For seat tests, valves shall be tightened by one person, tightening only to the closure torque recommended by the manufacturer (see 8.1.1.7). The torque is to be applied by a calibrated torque wrench with an attachment that is mounted over the center of the stem/shaft. If the torque recommended by the manufacturer should prove to be inadequate, the torque should be increased incrementally, to a maximum of 1.25 times the recommended value, until the seat leakage is within allowable limits. The required closure torques shall be measured and reported. During seat tests, external forces that affect seat leakage shall not be applied to the valve ends. The duration of the closure tests shall be 120 seconds for valves NPS 4 and smaller and 300 seconds for valves NPS 6 and larger.

8.1.2.3 The following dimensions and finishes shall be measured on each valve, as applicable, and compared with those specified in ASME and API standards, and the manufacturer's standards:

a. Face-to-face dimension (ASME B16.10).
b. Flange dimensions (ASME B16.5), including orientation of bolt holes.
c. Butt-welding end dimensions (ASME B16.25).
d. Center to top (closed and open position).
e. Handwheel diameter.
f. Facing finish, including number of grooves per inch, of raised-face end flanges (ASME B16.5) and bonnet-joint flanges.
g. Type of bolting threads (ASME B1.1 and B18.2.2).
h. Body and bonnet (or cover) thickness.
i. Stem diameter at the extremities and midpoint of the packing contact area, including minimum OD and thread major/minor diameters, surface finish, and straightness.
j. Wedge-wear travel (API Standard 600).
k. Stem projection (API Standard 600).
l. Number and size of bonnet or cover bolts.
m. Stuffing box dimensions and surface finish, and gland follower dimensions.
n. Socket-weld ends and threaded ends (ASME B16.11).
o. Gasket dimensions.
p. Port opening (bore and seat ring diameters).
q. Backseat ID and finish.
r. Bonnet joint dimensions.

8.1.2.4 All valve parts, especially the pressure-containing parts, shall be visually examined to determine the following, where applicable:

a. That the markings are as specified in the applicable product standard.
b. That the identification plate is as specified in the applicable product standard.

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c. That the construction is as specified in the applicable product standard.
d. That the handwheel operates clockwise to close when viewed from the outboard end of the stem, and that the handwheel/lever is properly marked with an arrow and the word OPEN to indicate the opening direction.
e. The packing type and arrangement, size, and number of rings as well as the number of rings that could be added after the shell hydrotest.
f. The method of attaching the seat ring to the body.
g. The presence of lubrication on the stem nut.
h. That grease or heavy lubricant has not been used on valve sealing surfaces (e.g., behind seat rings, on bonnet gasket surfaces, etc).
i. The type of closure element.
j. That the closure element does not tend to hang up in any orientation.
k. That the T-head is integral (without welding or weld build-up) with the stem.
l. That a fully open gate/plug/ball completely clears the seat openings.
m. That any disk nut on swing check valves is positively locked in place.
n. The number, location, and size of any tapped openings in pressure-containing parts.
o. The type and direction of the stem threads.
p. That a photograph of the disassembled valve parts, readable name plate, and cast-in markings has been made.
q. That the markings on the bolting for the body, bonnet, and cover joints are as specified in applicable ASTM specifications.
r. The method of attachment of the handwheel nut.
s. If tack welding is used in securing various components.
t. The type of end protection used in shipment.
u. That the body-to-bonnet/cover gasket design and material are as specified.

8.1.2.5 After the testing specified in 8.1.2.2 and the visual examinations specified in 8.1.2.4, paint and sealants shall be removed from the bodies, bonnets, and covers, and each valve pressure-retaining component shall be visually examined to determine the following:

a. That forgings are free from laps and seams.
b. That the surface quality of castings, including the body, bonnet, and cover, is as specified in MSS SP-55.

8.1.2.6 The material tests listed in Table 1 shall be made on a minimum of five of the sample valves, making sure that the source of each body and bonnet (or cover) as well as each material group from each material source is sampled. Chemical composition and hardness shall be nondestructively determined except that small samples may be removed from the body, bonnet, or cover in a manner that will not affect the integrity of the component (for example, areas such as bosses, ribs, and flange perimeters).

8.1.2.7 Strength tests of the stem/shaft-to-closure element connection shall be performed on valves in accordance with Appendix B. (See Appendix A for suggested selections.)

8.1.2.8 All pressure-retaining welds shall be completely radiographed in accordance with the requirements of Table 341.3.2A of ASME B31.3, using the acceptance criteria for normal service conditions. Butt-welding end preparations, welds in fabricated wedges, and pressure-retaining welds that could not be radiographed shall be examined in accordance with ASME B31.3 by either the magnetic particle or the liquid penetrant method.

Table 1—Material Tests

Tests	Body	Cover/ Bonnet	Seat Ring	Stem	Yoke Nut	Disk	Back Seat	RTJ Bonnet Gasket	Bonnet Bolts[c]
Chemical Composition	x	x	x[a]	x	x	x[a]	x	x	x
Tensile Strength	b	b	—	—	—	—	—	—	x[d]
Yield Strength	b	b	—	—	—	—	—	—	—
Elongation	b	b	—	—	—	—	—	—	—
Reduction of Area	b	b	—	—	—	—	—	—	—
Hardness (HB)	x	x	x	x	x	x	x	x	x

Note: RTJ = ring type joint.
[a]If seating surfaces are welded, chemical analysis shall be made on both metals (weld metal and base metal). If seating surfaces are applied in the form of thin plates welded to the disk, chemical analysis shall be made on the disk, the thin plates, and the attachment welds. Any welds attaching seats or seating surfaces shall be dye-penetrant examined. Neither cracks nor lack of fusion is allowed.
[b]These test results may be taken from mill test reports.
[c]Four sizes of bonnet bolts and nuts shall be tested, randomly selected from two of the largest and two of the smallest valve sizes in the sample lots from Table A-3.
[d]Bonnet bolt tensile strength shall be estimated using the measured hardness readings and the correlations (hardness to tensile strength) in ASTM A370.

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8.1.2.9 Sections of cast valves, as identified in paragraph 8.3.1.1 of ASME B16.34 and shown in Figures 6–14, from four valves or 25% of the sample lot (whichever is larger) shall be examined by radiography. A minimum of one valve from each foundry source shall be examined. The procedure shall be in accordance with ASME B16.34, Annex B. The laboratory shall report each type of discontinuity for each film, with sketches illustrating the locations of all films.

8.1.2.10 Four handwheels out of the sample lot shall be subjected to a hammer test. Using normal force, the hammer (3 lb. for valves NPS 4 and smaller, 10 lb. for valves NPS 6 and larger) should strike the outer rim between the spokes at an angle perpendicular to the plane of the handwheel, and any damage reported. In addition, each test handwheel shall be subjected to a torque test, applying three times the torque recommended by the manufacturer for closure. In applying the torque, the center of the handwheel is to be restrained and the force required to produce the desired torque applied to the outer rim of the handwheel at the spoke junction, using an attachment mounted to the wrench. This test is intended to demonstrate that the handwheels' material and design are adequate, and that they are not gray or white cast iron. If the handwheels are steel, it should be determined how they were made (i.e., stamping, pressing, casting, or welding). Any burrs or sharp edges shall be noted.

8.2 CERTIFICATION

As part of the test program, the manufacturer shall provide written certification, signed by an officer of the manufacturing company, which states that all valves furnished, regardless of size, pressure class, or materials of construction, are equivalent to the valves involved in the qualification procedure and comply with the applicable product standards.

8.3 SHOP SURVEYS

As part of the qualification, the purchaser or his representative may survey the valve manufacturer's shops, and, as the purchaser deems appropriate, may survey the facilities of primary suppliers to the valve manufacturer, such as foundries and forge shops. The valve manufacturer shall provide access upon request at times that are mutually agreed upon.

8.4 EVALUATION

The manufacturer shall assemble all the data required by this section, including the laboratory report required by 8.1.2.1, into a single submittal so that the purchaser can determine whether the valve manufacturer has met the purchaser's qualification requirements. The purchaser shall treat all data supplied under these requirements as proprietary to the valve manufacturer.

8.5 QUALIFICATION

Changes to any items listed below may void valves qualified to this recommended practice. The manufacturer shall notify the purchaser of any changes in the following:

a. Material grade.
b. Design that will affect the strength, operability, or flow characteristics of the valves.
c. Location of manufacturing facilities.
d. Suppliers of pressure-containing castings and forgings.
e. Quality control procedures.
f. Ownership.

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APPENDIX A—SUGGESTED SELECTION FOR EXAMINATIONS AND TESTS OF STEEL VALVES MADE IN ACCORDANCE WITH API STANDARDS AND ASME B16.34

The minimum suggested sample lot is provided in Tables A-1 through A-6.

The sample lot shall include at least one valve from each manufacturing plant, each body and bonnet (or cover) source, and each material group (carbon steel, low-chrome alloys, and stainless steels).

For each test valve, at least three similar valves should be available for selection. As noted in 8.1.2.1, if recommended valve sizes are not available for selection from manufacturer/distributor inventories, nor in sufficient quantities for selection, one size larger or smaller may be selected, or as agreed upon by the manufacturer and the user.

Table A-1—Minimum Suggested Sample Lot for Wafer and Wafer-Lug Check Valves Made in Accordance With API Standard 594

Quantity	NPS	Class
2	4, 18	150
2	3, 12	300

Note: NPS = Nominal pipe size.

Table A-2—Minimum Suggested Sample Lot for Metal Plug Valves Made in Accordance With API Standard 599

Quantity	NPS	Class	Strength Test (NPS)
2	4, 8	150	8
1	3	300	3
2	3, 6	600	—

Note: NPS = Nominal pipe size.

Table A-3—Minimum Suggested Sample Lot for Steel Valves Made in Accordance With API Standard 600/603 and ASME B16.34

Quantity	NPS	Class	Valve Type	Strength Test (NPS)
3	4, 12, 24[a]	150	Gate	4, 12
1	3	150	Globe	—
2	3, 12	150	Check	—
2	3, 12	300	Gate	3, 12
1	3	300	Globe	—
1	3	300	Check	—
2	3, 12[b]	600	Gate	—

Notes:
1. The applicable sections of API Standard 600 and 603 also apply to globe valves and check valves.
2. NPS = Nominal pipe size.
[a]NPS 24 or largest size for which approval is sought.
[b]NPS 12, Class 600, valves shall have butt-welding ends; all others shall have flanged ends.

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Table A-4—Minimum Suggested Sample Lot for Steel Valves Made in Accordance With API Standard 602

Quantity	NPS	Class	Valve Type	Ends	Strength Test (NPS)
3	3/4, 2, 4[a]	150	Gate, conventional	Flanged	4[a]
2	3/4, 2	800	Gate, conventional	Threaded	3/4
1	3/4	800	Globe	Threaded	—
1	3/4	800	Check	Threaded	—
2	3/4, 2	1500	Gate, conventional	Socket-welded	2
1	3/4	1500	Globe	Socket-welded	—
1	3/4	1500	Check	Socket-welded	—
2	3/4, 2	800	Gate, integrally reinforced extended body	One end butt-welded, one end threaded	3/4
1	3/4	1500	Gate, integrally reinforced extended body	One end butt-welded, one end threaded	—

Notes:
1. The applicable sections of API Standard 602 also apply to globe valves and check valves.
2. NPS = Nominal pipe size.
[a]NPS 4 or largest size for which approval is sought.

Table A-5—Minimum Suggested Sample Lot for Metal Ball Valves Made in Accordance With API Standard 608

Quantity	NPS	Class	Strength Test (NPS)
2	4, 8	150	4
2	3, 6	300	6

Note: NPS = Nominal pipe size.

Table A-6—Minimum Suggested Sample Lot for Butterfly Valves Made in Accordance With API Standard 609 (Category B)

Quantity	NPS	Class	Strength Test (NPS)
2	12, 24	150	12
2	4, 16	300	4

Note: NPS = Nominal pipe size.

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APPENDIX B—STRENGTH TESTS FOR STEM/SHAFT-TO-CLOSURE ELEMENT CONNECTIONS

B.1 Stem-to-Wedge Connection Tests for Steel Gate Valves

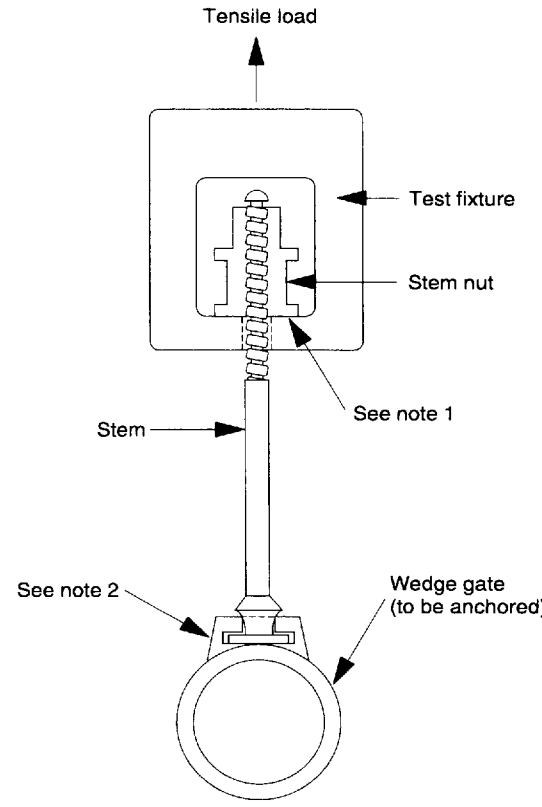
Direct tensile loads shall be applied to wedge-stem-stem nut assemblies as shown in Figure B-1 to determine the location of the first points of failure and the magnitude of the loads at failure. The first point of failure must occur at a location that is outside the valve pressure boundary.

If the first point of failure meets the requirements in the previous paragraph, the tensile testing shall continue to determine the failure load for the stem-to-wedge connection. A comparison shall be made between the first-point-of-failure load and the stem-to-wedge connection failure load.

If any of the tested assemblies fail to meet the requirements of the first paragraph, all of the gate valves in the sample lot shall be tested, and each failure shall be analyzed. If a failure is not due to design causes, then three additional assemblies of the same size as the original inadequate valve shall be tested.

A dimensional analysis of the stem-to-wedge connection of all of the valve sizes for which approval is sought shall demonstrate that all of the connections are similarly proportioned to the tested stems and wedges. If different product forms or different material mechanical properties are involved with untested valve sizes, additional stem-to-wedge tests shall be made to demonstrate the soundness of the other designs and/or materials.

If specified by the purchaser, an optional torque-to-failure test may be performed on API 602 valves. The parameters and acceptance criteria of such testing shall be mutually agreed upon by the manufacturer and purchaser. If such testing is used, the manufacturer shall provide calculations documenting theoretical torsional failure loads.



Notes:
1. The area of bearing shall not exceed the actual bearing area in the assembled valve.
2. Where the wedge design incorporates additional strength from the valve body, a device providing similar strength may be used to support the wedge during the performance of the stem-to-wedge connection test.

Figure B-1—Typical Stem-to-Wedge Connection Test Arrangement to Determine First Point of Failure

B.2 Stem-to-Plug or Stem-to-Ball Connection Tests for Metal Plug/Ball Valves

Direct torsional loads shall be applied to stem-to-plug/ball assemblies as shown in Figure B-2 to determine the location of the first points of failure and the magnitude of the loads at failure. The first point of failure must occur at a location that is outside the valve pressure boundary.

If the first point of failure meets the requirements in the previous paragraph, the testing shall continue to determine the failure load for the stem-to-plug/ball connection. A comparison shall be made between the first-point-of failure load and the stem-to-plug/ball connection failure load.

If any of the tested assemblies fail to meet the requirements of the first paragraph, all of the valves in the sample lot shall be tested, and each failure shall be analyzed. If a failure is not due to design deficiencies, then three additional assemblies of the same size as the inadequate valve shall be tested.

A dimensional analysis of the stem-to-plug/ball connection on all of the valve sizes for which approval is sought shall demonstrate that all the connections are similarly proportioned to the tested stems and plugs/balls. If different attachment methods or different material mechanical properties are involved with untested valve sizes, additional stem-to-plug/ball tests shall be made to demonstrate the soundness of the other designs and/or materials.

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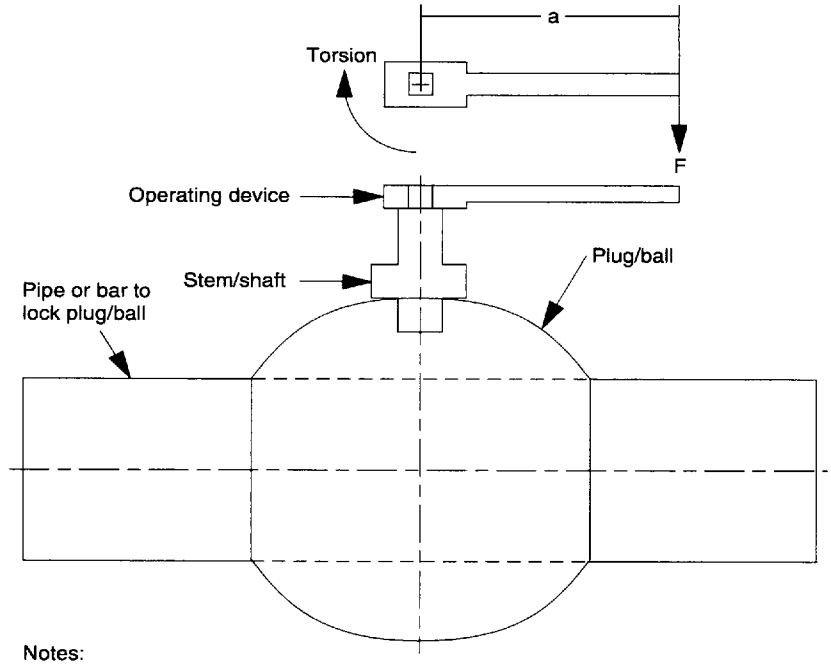


Figure B-2— Typical Stem-to-Plug/Ball Connection Test Arrangement to Determine First Point of Failure

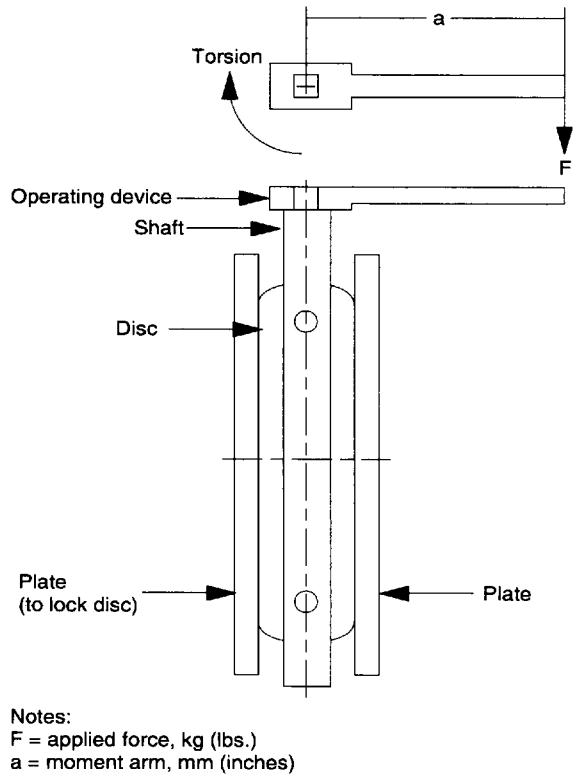


Figure B-3—Typical Shaft-to-Disc Connection Test Arrangement to Determine First Point of Failure

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B.3 Shaft-to-Disc Connection Tests for Butterfly Valves (Category B)

Direct torsional loads shall be applied to the shaft-disc assemblies as shown in Figure B-3 to determine the locations of the first points of failure and the loads at failure. The first point of failure must occur at a location that is outside the valve pressure boundary.

If the first point of failure meets the requirements above, the testing shall continue to determine the failure load for the shaft-to-disc connection. A comparison shall be made between the first-point-of-failure load and the shaft-to-disc connection failure load.

If any of the tested assemblies fail to meet the requirements of the first paragraph, all of the butterfly valves in the sample lot shall be tested, and each failure shall be analyzed. If a failure is not due to design deficiencies, then three additional assemblies of the same size as the original valve shall be tested.

A dimensional analysis of the shaft-to-disc connection on all of the valve sizes for which approval is sought shall demonstrate that all of the connections are similarly proportioned to the tested shafts and discs. If different attachment methods (spline, keys, pins, etc.) or different material mechanical properties are involved with untested valve sizes, additional shaft-to-disc tests shall be made to demonstrate the soundness of the other designs and/or materials.

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